

# Restricted Monomial Groups

Exact sigma–Molien formulas, proof, and matrix verification

Computational verification note

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**Outcome.** The general dual-code formula and the specialized  $G(r, p, n)$  coefficient formula were compared with explicit matrix averages in six representative cases. All 180 coefficients through total degree six agree. A direct Reynolds projector for  $G(2, 2, 3)$  on  $\text{Sym}^2(V) \otimes V$  has rank and trace 1, with idempotence error  $3.21 \times 10^{-16}$ .

## 1 Statement of the problem

Let  $G$  be a finite group and  $\rho : G \rightarrow GL(V)$  a complex representation. For independent degree markers  $\mathbf{t} = (t_1, t_2, \dots)$  define

$$\mathcal{S}_{G,V}(\mathbf{t}) = \frac{1}{|G|} \sum_{g \in G} \prod_{a \geq 1} \det(I - t_a \rho(g))^{-1}. \quad (1)$$

The coefficient of  $t_1^{\tau_1} \cdots t_\ell^{\tau_\ell}$  is claimed to be

$$\dim \left( \bigotimes_{a=1}^{\ell} \text{Sym}^{\tau_a} V \right)^G. \quad (2)$$

We prove this master identity, specialize it to code-constrained monomial groups, and independently test the resulting formulas on explicitly constructed matrices.

## 2 The master identity

Suppose  $g$  has eigenvalues  $\lambda_1, \dots, \lambda_n$ . The generating function for complete homogeneous symmetric polynomials gives

$$\det(I - tg)^{-1} = \prod_{i=1}^n (1 - t\lambda_i)^{-1} = \sum_{d \geq 0} h_d(\lambda_1, \dots, \lambda_n) t^d.$$

But  $h_d(\lambda_1, \dots, \lambda_n)$  is the character of  $\text{Sym}^d V$  at  $g$ . Consequently, the requested coefficient of (1) is

$$\frac{1}{|G|} \sum_{g \in G} \prod_a \chi_{\text{Sym}^{\tau_a} V}(g) = \text{tr} \left( \frac{1}{|G|} \sum_{g \in G} g \left| \bigotimes_a \text{Sym}^{\tau_a} V \right. \right).$$

The averaged operator is the Reynolds projector onto the invariant subspace. Its trace is its rank, proving (2). This proof also explains the three computational routes used below: determinant coefficients, character averages, and projector ranks must agree.

### 3 Code-constrained monomial groups

Fix  $\zeta = e^{2\pi i/r}$ , an  $H$ -stable additive code  $A \leq (\mathbb{Z}/r\mathbb{Z})^n$ , and

$$G(A, H) = D(A) \rtimes H, \quad D(A) = \{\text{diag}(\zeta^{x_1}, \dots, \zeta^{x_n}) : x \in A\}.$$

We use the convention  $g = D(x)P_\pi$ . If  $C$  is a cycle of  $\pi$ , the restriction of  $g$  to its coordinate subspace is a weighted cyclic shift. Only the full traversal contributes to its characteristic polynomial, hence

$$\det(I - tg) = \prod_{C \in \text{Cyc}(\pi)} \left(1 - \zeta^{x_C} t^{|C|}\right), \quad x_C = \sum_{i \in C} x_i. \quad (3)$$

Substitution in (1) proves the finite exact formula

$$\mathcal{S}_{A,H}(\mathbf{t}) = \frac{1}{|H||A|} \sum_{\pi \in H} \sum_{x \in A} \prod_{a, C} \left(1 - \zeta^{x_C} t_a^{|C|}\right)^{-1}. \quad (4)$$

#### 3.1 The dual-code filter

Expand each factor geometrically and let  $m_{C,a} \geq 0$  be its selected multiplicity. Put  $M_C = \sum_a m_{C,a}$  and set  $e_i = M_C$  for  $i \in C$ . The phase of a term is  $\zeta^{x \cdot e}$ . For

$$A^\perp = \{e \in (\mathbb{Z}/r\mathbb{Z})^n : x \cdot e = 0 \pmod{r} \text{ for all } x \in A\},$$

finite character orthogonality gives

$$\frac{1}{|A|} \sum_{x \in A} \zeta^{x \cdot e} = \mathbf{1}_{e \in A^\perp}.$$

Therefore the coefficient indexed by  $\tau = (\tau_1, \dots, \tau_\ell)$  is

$$\frac{1}{|H|} \sum_{\pi \in H} \# \left\{ (m_{C,a}) : \begin{array}{l} \sum_C |C| m_{C,a} = \tau_a \text{ for every } a, \\ e(\pi, m) \in A^\perp \end{array} \right\}. \quad (5)$$

Equation (5) is the implemented general formula. Notice that it contains no matrix eigenvalue computation and no average over diagonal matrices.

For a residue  $s \in \mathbb{Z}/r$ , the equivalent root-of-unity filter is

$$\Phi_{d,s}(\mathbf{t}) = \frac{1}{r} \sum_{\ell=0}^{r-1} \zeta^{-\ell s} \prod_a (1 - \zeta^\ell t_a^d)^{-1}.$$

It collects those monomials for which  $\sum_a m_a \equiv s \pmod{r}$ .

### 4 Specialization to $G(r, p, n)$

Assume  $p \mid r$  and

$$A_{r,p,n} = \{x \in (\mathbb{Z}/r\mathbb{Z})^n : \sum_i x_i = 0 \pmod{p}\}.$$

The vectors  $e_i - e_j \in A_{r,p,n}$  force a dual vector to have equal coordinates. The vectors  $pe_i \in A_{r,p,n}$  then force the common coordinate to be a multiple of  $r/p$ . Thus

$$A_{r,p,n}^\perp = \{q(r/p)(1, \dots, 1) : 0 \leq q < p\}. \quad (6)$$

The cycle-index theorem now yields

$$\mathcal{S}_{G(r,p,n)}(\mathbf{t}) = \sum_{q=0}^{p-1} [u^n] \exp \left( \sum_{d \geq 1} \frac{u^d}{d} \Phi_{d,qr/p}(\mathbf{t}) \right). \quad (7)$$

An equivalent coefficient algorithm regards each coordinate as carrying a label  $(m_1, \dots, m_\ell)$  whose coordinate sum has residue  $qr/p$ . An  $S_n$ -orbit is a multiset of  $n$  labels. Bounded dynamic programming over label multiplicities extracts exactly the coefficient of (7); this is the second independent formula route in the software.

With only  $t_1 = t$  nonzero, coefficient extraction reduces to

$$\mathcal{S}_{G(r,p,n)}(t, 0, \dots) = \frac{1}{(1 - t^{nr/p}) \prod_{j=1}^{n-1} (1 - t^{rj})}. \quad (8)$$

For  $W(D_n) = G(2, 2, n)$ , the second residue sector requires every coordinate to carry odd total multiplicity. It cannot occur below degree  $n$ . In particular, it does not create a degree-one fixed vector; this is the key zero-coordinate correction in the proposed discussion.

## 5 Independent verification design

1. Construct every matrix  $D(x)P_\pi$ . Check the predicted order  $r^n n! / p$  in the  $G(r, p, n)$  cases.
2. For every partition  $\tau$  of every total degree from 0 through 6, compute  $h_{\tau_a}$  recursively from the eigenvalues of each matrix and average  $\prod_a h_{\tau_a}$  over the group.
3. Independently evaluate (5) for a general code case or the multiset coefficient of (7) for  $G(r, p, n)$ .
4. For a selected target, explicitly embed normalized occupancy bases of symmetric powers in tensor powers, form the full Reynolds matrix, and compare its numerical rank and trace with the coefficient.

These routes share only the mathematical input  $(r, p, n)$  or  $(A, H)$ . The formula route does not call the matrix-average code.

## 6 Representative cases and results

There are 30 partitions through total degree six. The following suite includes an  $H = C_3$  code group not covered by the full symmetric-group specialization, the unrestricted wreath products, the  $D_n$  parity sector, and two different nontrivial divisors  $p$  of  $r$ .

| Case                         | $ G $ | $\dim V$ | degree | checks | result      |
|------------------------------|-------|----------|--------|--------|-------------|
| even-sign code $\rtimes C_3$ | 12    | 3        | 0–6    | 30     | <b>PASS</b> |
| $G(2, 1, 2)$                 | 8     | 2        | 0–6    | 30     | <b>PASS</b> |
| $G(2, 2, 3) = W(D_3)$        | 24    | 3        | 0–6    | 30     | <b>PASS</b> |
| $G(3, 1, 2)$                 | 18    | 2        | 0–6    | 30     | <b>PASS</b> |
| $G(4, 2, 2)$                 | 16    | 2        | 0–6    | 30     | <b>PASS</b> |
| $G(4, 4, 2)$                 | 8     | 2        | 0–6    | 30     | <b>PASS</b> |

Selected matched dimensions illustrate that the test is multivariate, not only the ordinary Molien specialization.

| Case                         | (2) | (3) | (2, 1) | (3, 2) |
|------------------------------|-----|-----|--------|--------|
| even-sign code $\rtimes C_3$ | 1   | 1   | 1      | 4      |
| $G(2, 1, 2)$                 | 1   | 0   | 0      | 0      |
| $G(2, 2, 3)$                 | 1   | 1   | 1      | 3      |
| $G(3, 1, 2)$                 | 0   | 1   | 1      | 0      |
| $G(4, 2, 2)$                 | 0   | 0   | 0      | 0      |
| $G(4, 4, 2)$                 | 1   | 0   | 0      | 0      |

For  $G(2, 2, 3)$  and  $\tau = (2, 1)$ , the target dimension is  $\binom{4}{2}\binom{3}{1} = 18$ . Direct averaging of the eighteen-dimensional target matrices produced

$$\text{rank } P = 1, \quad \text{tr } P = 0.9999999999999998, \quad \|P^2 - P\|_F = 3.21 \times 10^{-16}.$$

This agrees with both coefficient routes.

## 7 Reproducibility and interpretation

Run the command-line suite from the repository root with

```
.venv/bin/python -m for_this_guy.matrix_group_formula_verification.restricted_monomial_verification
```

or open `restricted_monomial_verification.py` with marimo. Matrix averages are accepted only when their imaginary part and distance from the nearest integer are below  $3 \times 10^{-7}$ . The observed residuals are much smaller; the displayed projector error is at machine precision.

Finite tests do not replace the preceding proof. Their role is to detect indexing errors, wrong permutation conventions, missing residue sectors, and incorrect treatment of zero-labeled coordinates. Within the stated small dimensions, no such discrepancy remains.